SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers sampled far beyond their training resolution need attention that is cheap and stays inside the training distribution. Recent training-free methods give every head the same hand-scheduled receptive field. We ask FLUX.1-dev, at its training resolution, how much of each head's output a native-size window explains and how much a downsampled view of the whole image explains. A per-head, per-noise-level convex mix of these two in-distribution branches fits full attention closely. Its weight is stable across prompts (split-half correlation 0.99) and transfers across emulated scales and calibration resolutions at almost no cost. Scale-Covariant Attention (SCA) reuses this table at the target resolution. For the first 8 of 28 steps, while only layout-scale structure carries signal, every head uses the global branch alone, whose keys are pooled onto the native grid with rescaled coordinates. Afterwards each head mixes the global branch with an exact native-extent sliding-window attention by its calibrated weight. SCA needs one calibration pass and no training, and its attention cost is linear in the token count. On 191 Aesthetic-4K prompts at 4096^2, under one protocol shared with AReA's released code, SCA raises ImageReward from 0.92 to 1.23 and HPSv2.1 from 0.272 to 0.297, lowers FID from 146.0 to 138.4, and is significantly better on every preference, alignment, and no-reference quality metric we report. At 71 s per image on one GH200 it is 2.6 times faster than full attention and 3.0 times faster than AReA. Both parts of the method are needed: without the opening phase subjects duplicate, and replacing the per-head mix with local-only attention after it lowers ImageReward by 0.36, because the mix is what carries global context through the remaining steps.

1 INTRODUCTION

Text-to-image diffusion transformers such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained at about one megapixel. Sampling them at 4096×4096 multiplies the number of image tokens by 16 and the cost of full attention by about 256. Speed is the smaller problem. Every attention layer then sees a token count and a range of relative positions outside its training distribution, and the outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025; Rajabi et al., 2026a).

Most training-free fixes edit either the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or the set of keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a) is the most recent of the second kind. For the first 8 of 28 steps, full-resolution queries attend to keys and values subsampled onto a native-size grid, with query coordinates divided by the pooling factor; afterwards, queries attend inside windows that shrink geometrically to the native size, with a wavelet split test, a halo ring, random shifts, and a global key-value bridge to repair windowing artifacts. Every head of every layer receives the same receptive field at a given step, and the schedule is set by hand.

We question the shared receptive field. AReA's evidence for it, the narrowing of attention during denoising, is an average over heads and layers. We ask the pretrained model instead: at its training resolution, where full attention is trustworthy, how much of each head's output does a native-size neighborhood explain, and how much does a downsampled view of the whole image explain? A convex mix of the two explains each head well (Section 4). The weight differs strongly between

heads, barely changes across prompts, and changes little with the emulated scale or the calibration resolution. We fit it once per model in closed form against the model's own full attention.

Output matching cannot set everything. Applied from the first step at 4096^2, the calibrated mix duplicates subjects: one prompt for a Moai statue gives five statues. At high noise only the lowest spatial frequencies carry signal, and these set layout and object count. A native window covers a quarter of the image width at 4096^2, cannot resolve those frequencies, and invents a layout per window. Squared output error does not register this, because texture dominates it. SCA therefore runs the global branch alone while $\sigma > 0.89$ (the first 8 of 28 steps) and the calibrated mix afterwards (Figure 1). The opening phase coincides with AReA's; the two methods differ in the remaining 20 steps.

The ablations separate the two parts. The opening phase is essential: without it, subjects repeat. After it, the calibration says that most heads lean local (mean weight 0.64 to 0.74), yet forcing every head to local-only attention lowers ImageReward by 0.36 and HPSv2.1 by 2.9 points. The minority weight each head puts on the global branch is what keeps global context in the last 20 steps. An earlier configuration with a low-band guard hid this, because the guard injects the global branch's low band into every head whatever its weight.

Our contributions are:

- A protocol that emulates resolution extrapolation inside the training distribution and scores any receptive-field rule by its output error against full attention, from statistics collected in one pass.
- Evidence on FLUX.1-dev that receptive-field needs are head-specific, prompt-stable, and close to invariant under changes of scale and of calibration resolution.
- SCA, a training-free attention processor: a global-only opening phase, then a calibrated per-head mix of a native-grid global branch and an exact FlexAttention sliding window. One attention layer at 4096^2 takes 18.7 ms against 87.3 ms for full attention.
- A 4K evaluation on Aesthetic-4K under one protocol shared with AReA's released code. On 191 prompts at 4096^2, SCA reaches ImageReward 1.23 against 0.92, FID 138.4 against 146.0, and HPSv2.1 0.297 against 0.272, and runs 2.6 times faster than full attention and 3.0 times faster than AReA on a GH200. AReA's paper reports ImageReward 1.32 under its own settings, so numbers across papers are not comparable. Ablations cover every component and the local-only and hard-routing special cases.

2 Related work

Training-free high-resolution generation. For U-Nets, ScaleCrafter dilates convolutions (He et al., 2023), HiDiffusion restructures the network and uses shifted windows (Zhang et al., 2023), FouriScale constrains the spectrum (Huang et al., 2024), and Jin et al. (2023) rescale the attention temperature. DemoFusion (Du et al., 2023), FAM (Yang et al., 2024), and FreeScale (Qiu et al., 2024) fuse a global path with local patches; I-Max (Du et al., 2024), HiFlow (Bu et al., 2025), and ScaleDiff (Koh et al., 2025) guide sampling with a native-resolution pass. Another line edits rotary embeddings (Su et al., 2021): Lumina-Next (Zhuo et al., 2024), DyPE (Issachar et al., 2025), UltraImage (Zhao et al., 2025), SigMa (Zhao et al., 2026), SEGA (Rajabi et al., 2026b), and TIDE (Liu et al., 2026). The closest relatives of SCA change which keys a query sees. ResDiT fuses a global branch with scaled positions and patch-wise attention by Gaussian and Fourier weights (Ma et al., 2025). HRDiT gives each head a static local scope chosen by an integer program under a cost budget (Xue et al., 2026). AReA schedules a shared receptive field from pooled-global to shrinking-local (Rajabi et al., 2026a). SCA differs from all three at once: its weights are per head and per noise level, its global branch is a zoom (pooling plus coordinate rescaling), and its weights are read off the model's own full attention instead of being set by hand or searched against a cost proxy.

Efficient and head-specific attention. ToDo attends from full-resolution queries to downsampled keys (Smith et al., 2024), the topology of our image-anchored branch without coordinate rescaling. DiTFastAttn selects window attention per layer and step (Yuan et al., 2024); for video, Sparse VideoGen classifies heads online (Xi et al., 2025), Sliding Tile Attention computes tiled windows

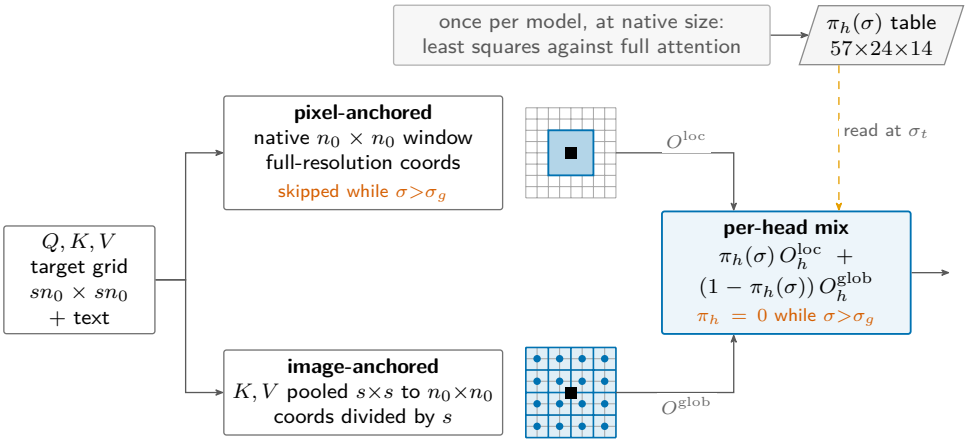


Figure 1: **SCA mixes two in-distribution receptive fields per head.** For a target grid of $sn_0 \times sn_0$ tokens, the pixel-anchored branch lets each query tile attend to a native $n_0 \times n_0$ key window with the original RoPE coordinates. The image-anchored branch subsamples keys and values by s onto an $n_0 \times n_0$ grid and divides query coordinates by s. Both see the native token count and range of relative positions. Head h mixes them with weight $\pi_h(\sigma)$ from a table calibrated once per model at native resolution. While $\sigma > \sigma_g$, $\pi_h(\sigma) = 0$ and the pixel-anchored branch is skipped. Sketches: black is a query, blue the keys it sees.

efficiently (Zhang et al., 2025b), and Sparse-vDiT searches per-head sparsity offline (Chen et al., 2025). These target speed at trained resolutions. Our local branch is blocked neighborhood attention (Hassani et al., 2022) written in FlexAttention (Dong et al., 2024). In language models, Self-Extend gives far tokens grouped positions (Jin et al., 2024), Ms-PoE rescales positions per head (Zhang et al., 2024), MoA profiles a span per head (Fu et al., 2024), DuoAttention splits heads into retrieval and streaming heads (Xiao et al., 2024; 2023), and YaRN interpolates RoPE frequencies (Peng et al., 2023). Trained vision transformers learn head-level scale splits: Focal attention (Yang et al., 2021), PVT (Wang et al., 2021), HiLo (Pan et al., 2022), Shunted attention (Ren et al., 2021), and Swin (Liu et al., 2021). Our measurements suggest that a full-attention diffusion transformer learns a comparable split implicitly, and SCA reads it out without training.

3 Scale-Covariant Attention

3.1 Setting

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens with $N = H \times W$ image tokens and applies joint attention in every block. FLUX.1 has 19 double-stream and 38 single-stream blocks with 24 heads each, and RoPE encodes each image token's grid position $p = (r, c)$. It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. For one head, full attention is

$$O_i^{\text{full}} = \sum_j \frac{\exp(\ell_{ij})}{\sum_{j'} \exp(\ell_{ij'})} v_j, \qquad \ell_{ij} = \tfrac{1}{\sqrt{d}} \left(R_{p_i} q_i\right)^\top \left(R_{p_j} k_j\right), \tag{1}$$

with R_p the rotation for position p. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer scale factors, written s when equal. The flow-matching latent at noise level $\sigma \in [0, 1]$ is $x_\sigma = (1 - \sigma)\, x_0 + \sigma\, \epsilon$. A softmax over any partition of the keys into sets A and B splits exactly into

$$O_i^{\text{full}} = m_i\, O_i^A + (1 - m_i)\, O_i^B, \tag{2}$$

where O_i^A, O_i^B are the outputs restricted to A and B and m_i is the softmax mass on A. With A a neighborhood of the query, every head is a query-dependent mix of a near field and a far field. SCA replaces both restricted outputs with branches that stay in distribution, and the mass m_i with a calibrated per-head weight.

3.2 TWO BRANCHES, TWO COORDINATE SYSTEMS

Pixel-anchored branch. The grid is cut into query tiles of $t \times t$ tokens ($t = 8$). Every query in a tile attends to all text keys and to the $n_0 \times n_0$ window of image keys centered on the tile and clamped to the grid, with the original full-resolution positions. Each query sees $n_0^2 + T$ keys, the native count, at relative offsets below n_0 per axis, the native range. This branch models content whose extent is fixed in tokens: edges, texture, fine structure. Its output is O_i^{loc}.

Image-anchored branch. Keys and values are subsampled at the center of every $s_H \times s_W$ block, giving n_0^2 pooled image keys. Pooled key b takes the block index as its position and query i takes $\lfloor p_i/s \rfloor$ per axis, so both share one native coordinate system. Text keys are not pooled. Each query again sees $n_0^2 + T$ keys at native offsets. This branch models content whose extent is a fixed fraction of the image: layout, object count, identity. Its output is O_i^{glob}. It is AReA's first phase with the pooling factor fixed at s (Rajabi et al., 2026a).

Per-head mixing with a noise gate. For image queries of head h at noise level σ,

$$O_i \;=\; \pi_h(\sigma)\, O_i^{\text{loc}} + \big(1 - \pi_h(\sigma)\big)\, O_i^{\text{glob}}, \qquad \pi_h(\sigma) = 0 \text{ for } \sigma > \sigma_g. \tag{3}$$

Text queries use the image-anchored branch alone. Relative to Equation 2, $\pi_h(\sigma)$ stands in for the near-field mass; it is constant across queries, which makes it cheap and lets a few prompts determine it.

Why the gate. At noise level σ, the signal-to-noise ratio of a spatial frequency band is $((1-\sigma)/\sigma)^2 S(f)$ for image spectrum S. Natural spectra fall with frequency, so in the first steps only the lowest band carries signal, and that band fixes layout and object count. A window of w tokens completes less than one period of any frequency below $1/w$ cycles per token, so it cannot tell one large object from a repeating pattern, and neighboring windows settle the question independently. Running the mix from the first step therefore duplicates subjects at the window's scale: with a 16-token window the duplicates shrink and multiply (Figure 4). The image-anchored branch sees the whole image on the native grid and resolves this band. SCA sets $\pi_h(\sigma) = 0$ and skips the pixel-anchored branch while $\sigma > \sigma_g = 0.89$, which covers the first 8 of 28 steps at the maximum shift. This equals AReA's switch point $\sigma^\star$. Gating 13 steps ($\sigma_g = 0.80$) produces a 4×4 block mosaic, because all queries in an $s \times s$ block share one coarse position in the image-anchored branch and receive nearly the same output at high noise. We did not measure the per-band SNR crossing on FLUX latents [TODO: measure per-band SNR of FLUX.1-dev latents against σ and mark where the band above $1/w$ crosses 1].

3.3 CALIBRATION BY OUTPUT MATCHING

The weights must come from where the model is trustworthy, which the target resolution is not. We emulate extrapolation inside the training distribution: at native resolution, an emulated scale s_c uses a local window of $(n_0/s_c)^2$ tokens, a pooling factor s_c onto an $(n_0/s_c)^2$ grid, and query tiles of t/s_c. The ratio of image to window is then s_c, as at a real target of scale s_c, while full attention at native resolution stays in distribution and serves as the reference. For every block, head, and noise-level bin we accumulate, over prompts, steps, image queries, and channels,

$$A = \sum \|O^{\text{loc}} - O^{\text{glob}}\|^2, \quad B = \sum \langle O^{\text{full}} - O^{\text{glob}},\, O^{\text{loc}} - O^{\text{glob}} \rangle, \quad E = \sum \|O^{\text{full}} - O^{\text{glob}}\|^2. \tag{4}$$

The squared error of the mix is $E - 2\pi B + \pi^2 A$, so the optimal weight is $\pi_h(\sigma) = \text{clip}(B/A, 0, 1)$. The same statistics score any other rule (per-layer weights, one shared weight per noise level, a hard switch at any threshold, local only, global only) without another forward pass. We use 14 uniform bins in σ and interpolate linearly between bin centers; the table has $57 \times 24 \times 14$ entries. Calibration runs the unmodified model at 1024^2, collects emulated scales 2 and 4 in one pass, and took [TODO: GPU hours] on one GH200 for 32 prompts (Appendix A).

3.4 INFERENCE AND COST

At a target with factors (s_H, s_W), SCA uses the table of the emulated scale closest in log space to $\max(s_H, s_W)$; all our targets use the scale-4 table. Algorithm 1 gives one layer. Weights, scheduler,

Algorithm 1 One SCA attention layer at noise level σ (per head h).

Require: pre-RoPE Q, K, V for T text and $N = H \times W$ image tokens; factors (s_H, s_W); table $\pi_h(\sigma)$; gate $\sigma_g = 0.89$

$\tilde{K}, \tilde{V} \leftarrow$ image keys and values at the center of every $s_H \times s_W$ block, plus all text keys

$O^{\text{glob}} \leftarrow \text{Attn}\big(Q \text{ at } \lfloor p/s \rfloor,\ \tilde{K} \text{ at block index},\ \tilde{V}\big)$ ▷ all queries, text included

if $\sigma > \sigma_g$ **then return** O^{glob} ▷ opening phase: pixel-anchored branch skipped

$O^{\text{loc}} \leftarrow$ FlexAttention of each $t \times t$ image-query tile over text keys and its $n_0 \times n_0$ window, original RoPE

return $[\, O^{\text{glob}}_{\text{text}},\ \pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}_{\text{img}} \,]$ ▷ mix in fp32

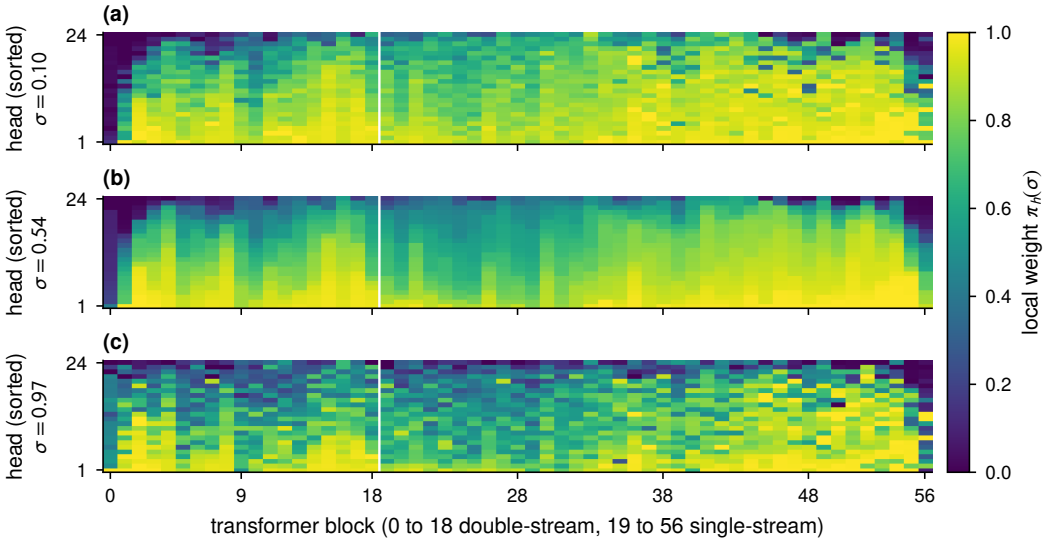


Figure 2: **Within every block, heads range from almost purely global to almost purely local.** Fitted $\pi_h(\sigma)$ for all 57 blocks and 24 heads of FLUX.1-dev at three noise levels, emulated scale 2, 32 prompts. Within each block, heads are sorted by their weight at $\sigma = 0.54$, in the same order in all panels. The white line separates double-stream from single-stream blocks.

VAE, and text encoders are untouched; the method is an attention-processor swap. Per head and block, full attention costs $(N + T)^2$ query-key pairs; SCA costs $N(n_0^2 + T)$ for the pixel-anchored branch, skipped in 8 of 28 steps, and $(N + T)(n_0^2 + T)$ for the image-anchored branch, both linear in N. At 4096^2 this is 8.4 times fewer attention FLOPs than full attention (Appendix B). The pixel-anchored branch is one FlexAttention call (Dong et al., 2024) whose block mask is the window membership itself, so it is an exact sliding window rather than an approximation.

4 What the heads want

We run the calibration on FLUX.1-dev with 32 prompts at 1024^2, guidance 4.5, 28 steps, and the shift μ at its maximum (the setting of Section 5), and repeat it with 16 prompts at 1280^2. The error measure is the relative output error $\sum \|O^{\text{full}} - O\|^2 / \sum \|O^{\text{full}}\|^2$ over image queries, pooled over blocks, heads, and steps.

Heads differ. At emulated scale 2 the standard deviation of $\pi_h(\sigma)$ over the 1368 heads is 0.24 to 0.26 at every noise level (Figure 2). Half of all (head, noise level) cells (50.4%; 49.7% at scale 4) lie in $[0.2, 0.8]$, 11.7% are above 0.95, and 2.9% below 0.05. The mean weight is 0.64 to 0.74 depending on the noise level, so the typical head leans local but still draws on the global branch. The spread is mostly within blocks: a per-layer table recovers only part of the gain of the per-head table (Figure 3). The median head changes its weight by 0.18 across noise levels, and a quarter of heads by more than 0.3.

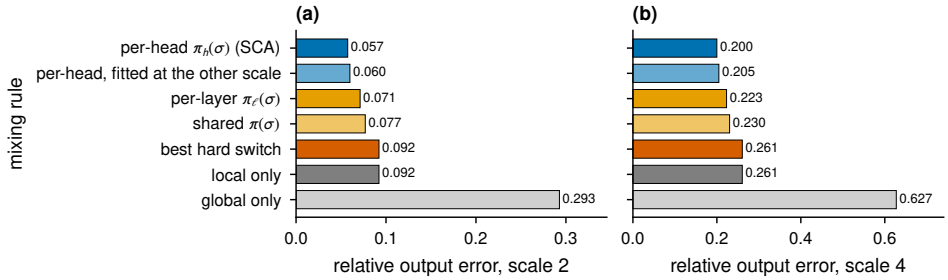


Figure 3: **Per-head weights explain full attention best, and transfer across scales.** Relative output error against full attention for each mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts. "Fitted at the other scale" applies the scale-4 table at scale 2 and vice versa. The best hard switch puts its threshold at $\sigma^\star = 1$ and so equals local only. In-sample and held-out errors agree within 0.0003.

The table is a property of the model, not of the prompts. Tables fitted on two disjoint halves of the prompts correlate at 0.989 over all cells (0.986 at scale 4). Each half's table scored on the other half gives a relative error of 0.0576, against 0.0574 in sample. At 1280^2, 16 prompts give a split-half correlation of 0.974.

The weights barely depend on scale or calibration resolution. The table fitted at emulated scale 2 and applied at scale 4 gives a relative error of 0.205 against 0.200 for the scale-4 table: 2.4% higher, and still below every shared rule fitted at scale 4 (Figure 3). The scale-4 table at scale 2 gives 0.0599 against 0.0574. Across calibration resolutions, the 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585, and the reverse gives 0.0578 against 0.0574. The two resolutions emulate scale 2 with windows of 32 and 40 tokens, so the weights depend on the ratio of image to window rather than on the window's absolute size. This is what we mean by scale covariance: as the image grows, each branch follows its own law and the weight between them stays put.

Output error and what it misses. Per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4, relative to the best shared hard switch; a per-layer table reaches 0.071 and 0.223, one shared weight per noise level 0.077 and 0.230. In output energy there is no global phase: even in the noisiest bin the global branch alone has error 0.27 at scale 2, against 0.14 for the local branch alone, and the best hard switch never switches (Appendix A, Figure 5). The gate of Section 3 contradicts this on purpose. Output error is dominated by texture and does not weigh the low band that sets layout at high noise; Section 5.3 shows that generation needs both: without the gate subjects repeat, and without the calibrated mix after it preference and alignment scores drop.

5 Experiments

5.1 Protocol

Model, sampler, baselines. FLUX.1-dev (Black Forest Labs, 2024) in bf16, 28 flow-matching Euler steps, guidance 4.5, and the timestep shift μ at its maximum (1.15) at every resolution, with one fixed seed per prompt shared across methods and tiled VAE decoding for all. These are AReA's FLUX.1 defaults, applied to every method so that only attention differs. Baselines are full attention at the target resolution (Base) and AReA (Rajabi et al., 2026a) from its released code under this protocol, with the FLUX.1-dev checkpoint in place of the FLUX.1-Krea-dev default of the release. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our protocol gives 0.92, so numbers across papers are not comparable (Appendix D). [TODO: decide whether DyPE, SEGA, UltraImage, and HRDiT are run.]

Benchmark. The 195 evaluation prompts of Aesthetic-4K with their 4K reference images (Zhang et al., 2025a), at 4096×4096, 2048×4096, and 2048×6144 (height $\times$ width)[1]. Four of the

[1] [TODO: confirm AReA's orientation convention and match it.]

Table 1: **Aesthetic-4K with FLUX.1-dev**, 191 prompts (development prompts excluded), guidance 4.5, 28 steps, one seed per prompt shared across methods. HPS, CLIP-IQA, and ARNIQA are $\times 100$. Higher is better except latency and FID. Bold: best per column. *: significantly better than AReA (95% paired bootstrap interval of the difference excludes zero). Latency is a clean single-GPU measurement, seconds per image on one otherwise idle GH200 at batch 1, not campaign wall-clock (Appendix B). SCA + band guard is the configuration of our first full run (Appendix C).

Method	Lat. (s)↓	IR	Aesth.	CLIP	PickS.	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096													
Full attention	184.5	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
AReA (released code)	215	0.92	6.28	27.67	22.35	28.64	27.20	146.0	110.2	39.4	50.4	67.5	67.3
SCA + band guard	88	1.20*	**6.46***	28.87*	22.90*	29.40*	28.95*	142.3	**104.8***	41.4*	61.0*	70.9*	70.8*
SCA (final)	**71**	**1.23***	6.45*	**29.11***	**23.09***	**29.77***	**29.71***	**138.4***	106.5	**45.3***	**61.2***	**72.9***	**71.2***
2048×4096: full, AReA, SCA	[TODO: runs in progress]												
2048×6144: full, AReA, SCA	[TODO: runs in progress]												

195 prompts (boats, Moai, woman and bear, underwater surfer) were used to choose the gate and window, so the main table reports the other 191. Ablations use a fixed subset of 46 of these.

SCA settings. $n_0 = 64$, tile $t = 8$, local window 64×64, block-center pooling, gate $\sigma_g = 0.89$, and the scale-4 table from the 32-prompt calibration at 1024^2. The calibration prompts are written for this purpose [TODO: verify that they are disjoint from Aesthetic-4K]. "SCA + band guard" additionally replaces the low band of each head's output by that of the image-anchored branch (Appendix C); it is the configuration of our first full run and not part of the final method.

Metrics. ImageReward (IR) (Xu et al., 2023), LAION aesthetic score (Schuhmann, 2022), CLIP score (ViT-L/14) (Radford et al., 2021), PickScore (Kirstain et al., 2023), HPSv2 and HPSv2.1 (Wu et al., 2023); FID (Heusel et al., 2017) against the 4K references at 299^2 and patch FID (FID_p) on four fixed 1024^2 crops per image; MUSIQ (Ke et al., 2021) on the resized image and on a 1024^2 crop (MUSIQ_c), CLIP-IQA (Wang et al., 2022), and ARNIQA (Agnolucci et al., 2024). Intervals are 95% percentile paired bootstraps over prompts with 1000 draws. Latency is wall-clock per image on one idle GH200 (96 GB), batch 1, including text encoding and VAE decoding, after a warm-up generation.

5.2 Main results

On 191 prompts at 4096^2, SCA beats AReA on every preference, alignment, and no-reference quality metric (Table 1). The paired differences are ImageReward $+0.31$ $[0.24, 0.38]$, aesthetic score $+0.17$ $[0.12, 0.22]$, HPSv2 $+1.12$ $[0.95, 1.31]$, HPSv2.1 $+2.50$ $[2.17, 2.88]$, PickScore $+0.73$ $[0.62, 0.86]$, CLIP score $+1.45$ $[1.06, 1.82]$, MUSIQ $+5.8$ $[4.1, 7.5]$, MUSIQ_c $+10.9$ $[9.5, 12.4]$, CLIP-IQA $+5.4$ $[4.0, 6.8]$, and ARNIQA $+3.9$ $[3.0, 4.9]$. FID falls by 7.6 $[3.5, 12.5]$ and patch KID falls significantly; patch FID falls by 3.7 $[-1.1, 7.2]$, within noise, as do KID and Inception score (6.52 vs. 6.42). At 71 s per image on one idle GH200, SCA is 2.6 times faster than full attention (184.5 s) and 3.0 times faster than AReA's released code (215 s). The band-guard configuration also beats AReA significantly on every metric in the table except FID, but scores below final SCA on ImageReward, CLIP score, PickScore, both HPS versions, and all four no-reference metrics, and takes 17 s longer per image; its patch FID is slightly lower. Per-layer, one attention layer at 4096^2 takes 18.7 ms against 87.3 ms for full attention (Appendix B). [TODO: qualitative figure: one prompt per resolution, all methods, zoomed crops.]

5.3 Ablations

Table 2 changes one component at a time. Panel (a) ablates the final method; panel (b) holds the full set of ablations we ran earlier around the band-guard configuration, in which every row except "no band" and v0 keeps the guard. All rows except v0 keep the gate.

The per-head mix supplies global context after the gate. Setting $\pi = 1$ after the gate, so that every head attends only inside its native window for the last 20 steps, lowers ImageReward by 0.36 $[0.22, 0.53]$, HPSv2.1 by 2.86 $[2.36, 3.36]$, PickScore by 1.20 $[0.99, 1.44]$, CLIP score by 1.93

Table 2: **Ablations at** 4096^2, 46 Aesthetic-4K prompts. Each panel starts with a reference row of absolute values; every other row changes one thing and reports its paired difference to that panel's reference. (a) Final setting: reference SCA final (gate 0.89, per-head $\pi_h(\sigma)$, no band guard). (b) Earlier setting: reference SCA + band guard (band 16). **Bold blue**: significantly better; *orange italic*: significantly worse (95% paired bootstrap CI excludes zero). The patch FID difference of the $\pi = 1$ row in (a) has no interval and is unmarked. HPS, CLIP-IQA, and ARNIQA are $\times 100$. "Mix err." is the relative output error of the weight rule at emulated scale 4 (Section 4).

Variant	Mix err.	IR	Aesth.	HPSv2	HPSv2.1	PickS.	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_c	$\text{FID}_p \downarrow$
(a) Final setting, no band guard												
SCA final (reference)	0.200	1.12	6.45	29.71	29.98	22.96	29.41	71.3	70.5	46.1	62.5	170.2
w/o per-head mixing ($\pi = 1$)	0.261	−0.36	−0.21	−1.78	−2.86	−1.20	−1.93	−4.6	−1.0	−3.5	−0.3	+24.1
one shared $\pi(\sigma)$	0.230						[TODO: running]					
per-layer $\pi_\ell(\sigma)$	0.223						[TODO: running]					
mean pooling	n/a						[TODO: running]					
(b) Earlier setting, with band guard												
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	169.3
no band (**= SCA final**)	0.200	−0.04	+0.01	+0.54	+1.17	+0.25	+0.40	+1.7	+0.9	+2.9	+0.4	+0.8
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+9.0
v0: no gate, no band	0.200	−1.37	−0.07	−1.35	−2.90	−1.94	−1.25	+4.6	+5.1	−2.7	+2.9	+35.5
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.8
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.4
local only ($\pi = 1$)	0.261	−0.01	+0.03	+0.16	+0.55	+0.12	+0.16	+0.5	+0.6	+1.0	+0.9	+1.3
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	+0.34	+0.09	+0.13	+0.6	+0.6	+0.8	+0.8	+1.1
global only ($\pi = 0$)	0.627	−0.44	−0.75	−2.79	−4.48	−1.19	−3.08	−25.2	−44.3	−0.6	−11.3	+69.5
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.8

$[1.21, 2.63]$, aesthetic score by 0.21, CLIP-IQA by 4.6, and MUSIQ by 3.5, all significant, and raises patch FID from 170.2 to 194.3; only ARNIQA and MUSIQ_c stay within noise. Most heads lean local (mean weight 0.64 to 0.74), but the calibrated share each head gives the image-anchored branch is what keeps the image coherent once the opening phase ends. In panel (b), local only matched the per-head mix within 2% on preference metrics and was even slightly higher on HPSv2.1 (+0.55). That came from the guard: it replaces the low band of every head's output with that of the image-anchored branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$. Without the guard the mix is the only route for global information after the gate.

The opening phase is essential. Removing the gate lowers ImageReward by 0.36 and every preference and alignment score significantly, and raises patch FID by 9.0. Without gate and band (v0, the calibrated mix from the first step) ImageReward falls by 1.37: subjects repeat across the image (Figure 4). v0 scores higher on CLIP-IQA and ARNIQA because each repeated subject is sharp; no-reference quality metrics do not see layout.

The global branch cannot work alone. Global only loses on every metric (patch FID +69.5, ARNIQA −44): with keys s tokens apart it cannot render detail. Mean pooling in place of block-center subsampling blurs the keys of the opening phase and lowers ImageReward by 0.85, confirming AReA's finding.

Shared and per-layer weights. With the guard, one shared weight per noise level and per-layer weights match the per-head table on preference metrics but lose significantly on CLIP-IQA, ARNIQA, and MUSIQ_c (−6.5, −4.6, −5.7 for the shared weight; −1.2, −1.2, −1.5 per layer); the shared weight has a lower patch FID (−4.8). Hard per-head routing (10 to 17% of heads global at threshold 0.5, about 6% at 0.3) also stays within 2% of the mix on preference metrics. Because the guard supplies global low frequencies by itself, these rows understate what the weights do in the final method; their no-band counterparts are in panel (a). Removing the band guard improves the per-head mix on five metrics and is our final method (Appendix C).

[TODO: image grid, rows Moai, boats, woman and bear, surfer; columns (a) v0: mix from the first step, 64×64 window; (b) v0 with a 16×16 window; (c) SCA final; (d) AReA; one zoomed crop per row.]

Figure 4: **Without the opening phase, the calibrated mix duplicates subjects at the window's scale.** 4096^2, same seed per row. (a) Five Moai statues; a sea of small boats. (b) A 16-token window gives smaller, more numerous copies. (c) SCA: one main subject. (d) AReA.

6 Limitations

What the per-head weights add. Without the band guard, the per-head mix clearly beats local-only attention after the gate. Whether it also beats one shared or per-layer weight in that setting is still being measured; with the guard, those variants matched it on preference metrics and lost only on no-reference quality. **Residual duplication.** With the gate alone, small extra figures can appear late in sampling on some prompts, and the underwater surfer prompt keeps doubled limbs in every variant we tried; AReA's image for that prompt shows a large translucent ghost. The band guard of Appendix C removes some of these cases at a cost in metrics and time. **The gate threshold** equals AReA's $\sigma^\star$, was compared only with 0.80 on four prompts, and its dependence on the shift and step count is untested. **Calibration objective.** Squared output error does not see low-band errors, which is why the gate is set by hand; a frequency-weighted calibration could fit it. **Emulation.** Emulated scale 4 at 1024^2 uses a 16-token window, the real 4096^2 target a 64-token window. Transfer across emulated scales and calibration resolutions supports the assumption that the weights depend on the ratio, but cannot prove it at the target. **Query-independent weights.** Equation 2 has a query-dependent weight; merging the branches through their log-sum-exp normalizers with a calibrated per-head bias would keep it at no extra attention cost; we have not tested this. **Scope.** FLUX.1-dev only [TODO: add a second MM-DiT or state the restriction in the claims].

7 Conclusion

At its training resolution, FLUX.1-dev shows how each attention head splits between a neighborhood fixed in tokens and a view of the whole image. The split differs between heads, holds across prompts, and changes little with scale, so it can be measured once and reused at 4K. Output matching sets the weights but not the opening phase: while only layout carries signal, every head must see the whole image. With that phase, SCA reaches ImageReward 1.23 against 0.92 for AReA's released code at 4096^2 under a shared protocol, FID 138.4 against 146.0, and HPSv2.1 0.297 against 0.272, while running 2.6 times faster than full attention and 3.0 times faster than AReA. Most heads lean local after layout forms, but the small calibrated share of the global branch still matters: local-only attention loses on nearly every metric.

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A CALIBRATION DETAILS

Statistics. For every emulated scale, block, head, and noise-level bin, the pass stores A, B, E, the error of the local branch alone, $\sum \|O^{\text{full}}\|^2$, and the query count, in float64, summed over image queries and channels. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the image-anchored branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 pooling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 pooling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window and 2×2 pooling onto a 40×40 grid.

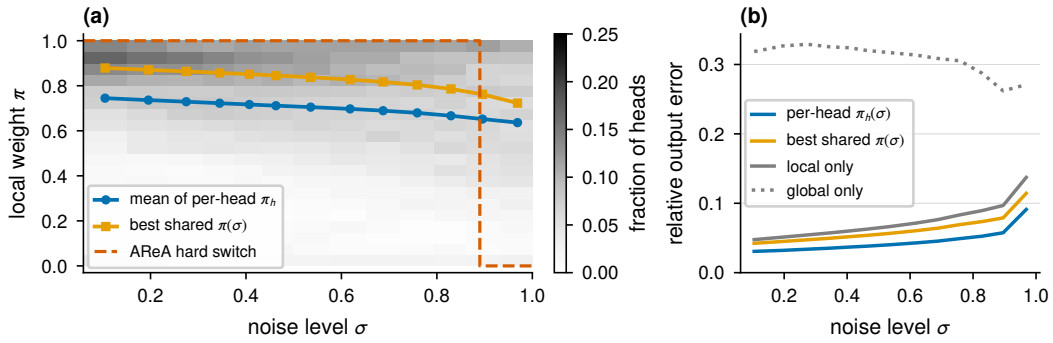


Figure 5: **In output energy, no noise level prefers the global branch.** (a) Distribution of $\pi_h(\sigma)$ over the 1368 heads per noise-level bin (grey), the mean over heads, and the best single shared weight, at emulated scale 2. The dashed step is AReA's schedule read as a weight. (b) Relative output error per bin. The local branch alone beats the global branch alone in every bin, and per-head mixing beats both. 32 prompts, one calibration run.

Noise-level bins. Bins have width $1/14$. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest bin is empty and copies its neighbor; 13 bins are populated. Bin centers are the mean σ of their steps. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure 5). The global share rises with noise, but at no noise level does the average head prefer the global branch. Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 5.3 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighborhood still receive 29% and 41%. A purely local receptive field cannot account for any step even at native resolution.

B Implementation details and cost

FlexAttention window. Image tokens are reordered tile-major; query blocks cover 8×16 tokens and key blocks 8×8, shifted so that unclamped windows start on a block edge, and the mask function is the window membership itself. The kernel matches a reference that gathers each tile's key window explicitly to within one bf16 unit in the last place, and both are equally far from an fp32 reference. RoPE, pooling, both branches, and the mix compile into one graph per layer.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256, so positions beyond 256 tokens per axis are rounded. This affects every method, including full attention and AReA, at the 2048×6144 setting (384 tokens per row). SCA takes the grid size from the sampler instead of inferring it from the rounded ids, so that its window and pooling geometry match the true grid.

End-to-end latency. On one idle GH200, batch 1, 4096^2, 28 steps, averaged over 3 prompts after warm-up: full attention 184.5 s, AReA (released code) 215 s, SCA + band guard 88.0 s (single-pass

Table 3: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the pixel-anchored branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Resolution ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
384×128	6144×2048	49.6	14.0	3.5

compiled band; outputs equal to the campaign code up to floating-point rounding). SCA final 71 s (median over three prompts of 71.0, 70.9, and 83.1 s). On the shared campaign nodes, per-image times were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (48 s) than in the clean measurement (17 s). Non-square resolutions: TODO.

C THE LOW-BAND DUPLICATION GUARD

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\mathrm{HP}_p = \mathrm{I} - \mathrm{LP}_p$. The guard replaces the low band of the mix M by that of the image-anchored branch:

$$O \;=\; \mathrm{LP}_p\big(O^{\mathrm{glob}}\big) + \mathrm{HP}_p\big(M\big) \;=\; O^{\mathrm{glob}} + \pi_h(\sigma)\,\mathrm{HP}_p\big(O^{\mathrm{loc}} - O^{\mathrm{glob}}\big). \tag{5}$$

The pixel-anchored branch then contributes only its difference from the image-anchored branch above the cutoff. The cutoff should lie below $1/(2s)$, where the image-anchored branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. The construction resembles FreeScale's fusion of a global low band with a local high band (Qiu et al., 2024) [TODO: check FreeScale's exact fusion rule and state the difference], applied here to per-head attention outputs in every layer.

What it fixes and what it costs. On the development prompts, the band alone (no gate) fixes the boats but leaves extra statues and faces on the Moai and woman-and-bear prompts; the gate alone gets the main subject right but small extra figures appear late on some prompts; gate and band together give one subject on the boats, Moai, and woman-and-bear prompts. On 46 prompts, however, removing the guard improves HPSv2 (+0.54), HPSv2.1 (+1.17), PickScore (+0.25), CLIP-IQA (+1.7), and MUSIQ (+2.9), all significant, leaves ImageReward, aesthetic score, ARNIQA, MUSIQ_c, and patch FID within noise, and saves 17 s per image on an idle GPU. The guard also hides what the per-head mix contributes: with it, local-only attention matches the mix, and without it local only loses on nearly every metric (Section 5.3). The guard is therefore an option for prompts where late duplication is visible, not part of the default method.

D REPRODUCING AREA

We run AReA's released code with its FLUX.1 defaults (block-center pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our shared protocol gives 0.92, so cross-paper numbers are not comparable, while all numbers in this paper come from one protocol. AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; we measure 215 s and 184.5 s on an idle GH200. [TODO: list the AReA commit, its schedules at the three resolutions, and any deviation from the paper, such as the cost cap on halo and bridge tokens.]